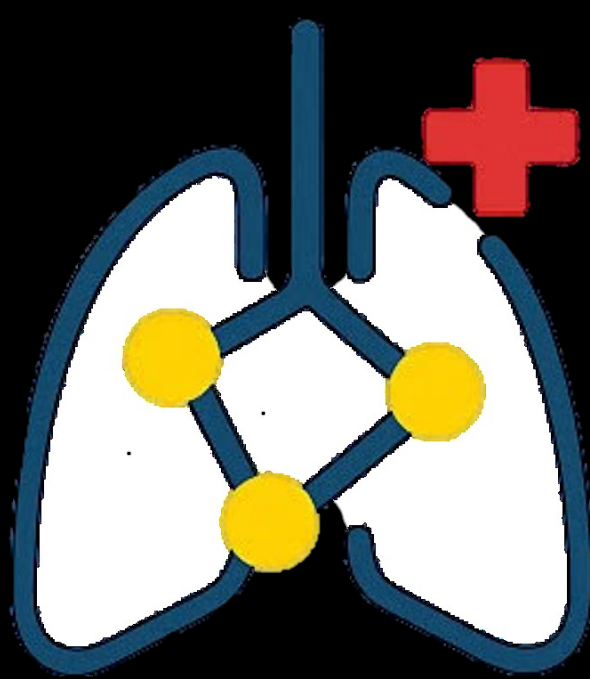




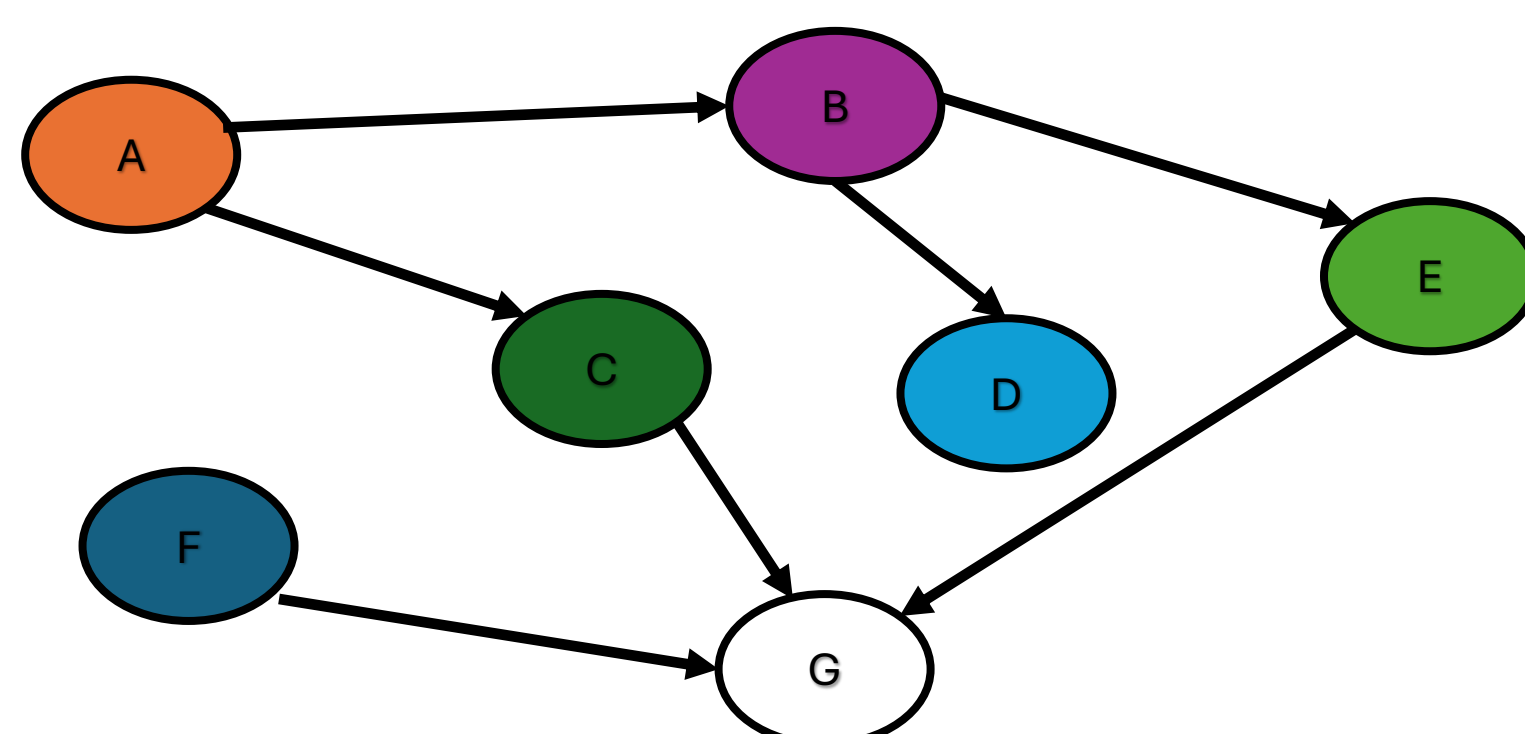
Comparative Analysis of Hybrid Structure Learning Algorithms for Bayesian Networks on Large-Scale Datasets

Gwangseong Cho, McClain Lincoln, Luciano Silveria, Shahir Tahiri, Aaron Taylor, Lourdes Juan

INTRODUCTION

Bayesian Belief Networks (BBNs) are **probabilistic graphical** models used to learn dependency structures from clinical data. These **structures** support **interpretable reasoning** for treatment planning and **outcome** prediction.

Because predictive **performance** depends heavily on the **quality of the learned graph**, this work evaluates hybrid structure-learning algorithms to determine which produces the most **accurate** and **clinically meaningful representations** in **large-scale** lung cancer datasets



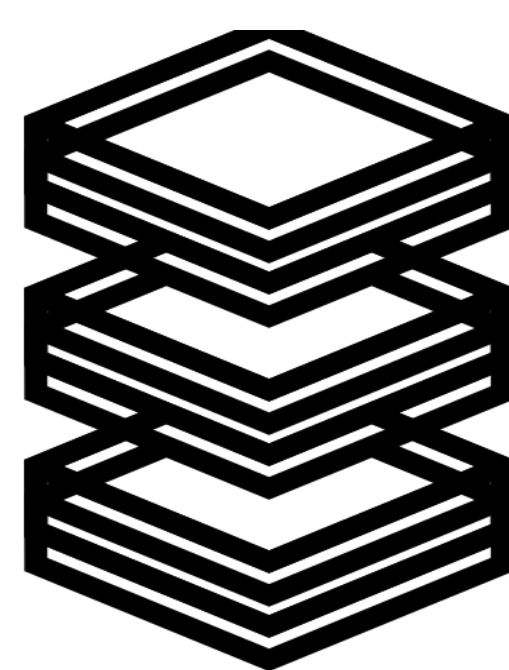
QUESTIONS

Which hybrid structure-learning algorithm provides the **best balance** of model fit, computational efficiency, structural interpretability, and **predictive performance** for large scale categorical lung cancer datasets?

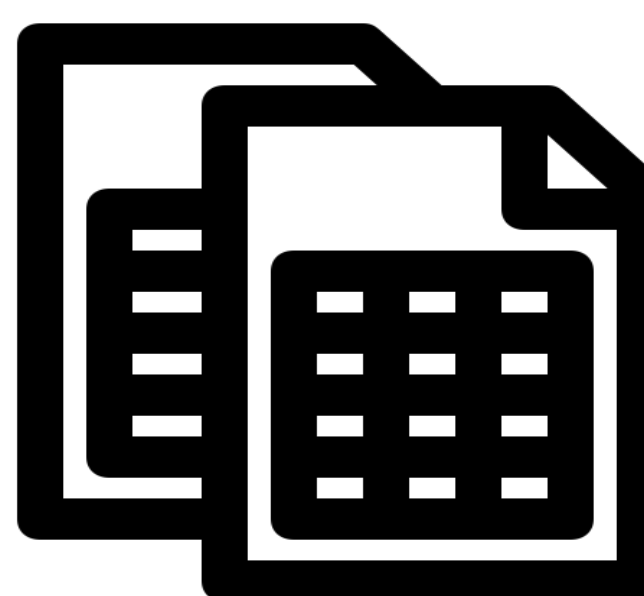
ANSWER

We compared MMHC, H2PC, and ARGES-like methods using BIC, runtime, edge count, and predictive accuracy on the same train/test splits. H2PC performed best overall, balancing strong accuracy, reasonable runtime, and a meaningful network structure. MMHC produced overly sparse graphs, while ARGES-like failed on categorical data

DATASET DETAILS: LUNG CANCER DATA



Massive lung cancer dataset
191,993 records
(80% train, 20% test)



11 Key Categorical Variables:
Age, Tumor Location, Cancer Cell Type, Treatment Plan, Extent of Regional Lymph Node Surgery, and Cause of Death

RICHNESS of DEPENDENCIES

H2PC identified **22 clinically meaningful dependencies**, including relationships such as Cancer Cell Type → Cause of Death. In contrast, **MMHC** identified only **7 dependencies**, limiting its ability to capture complex clinical interactions

CATEGORICAL vs. CONTINUOUS

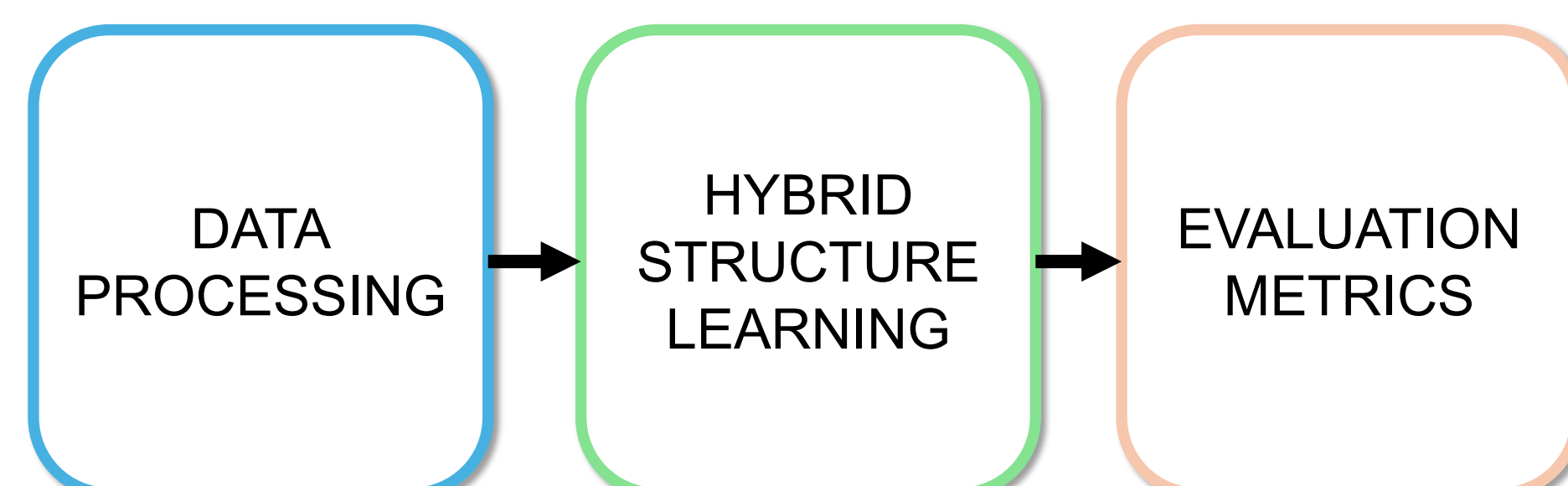
ARGES-like performed poorly on categorical data, suggesting it is better suited for **continuous-variable datasets**.

This highlights the importance of selecting structure learning methods based on data type

OPTIMAL BALANCE

H2PC provides the **best balance** between computational efficiency, **structural richness**, and **predictive performance**, making it the most suitable method for this dataset.

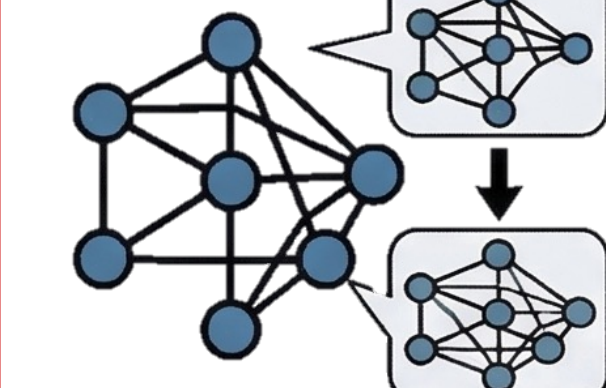
EXPERIMENTAL FRAMEWORK



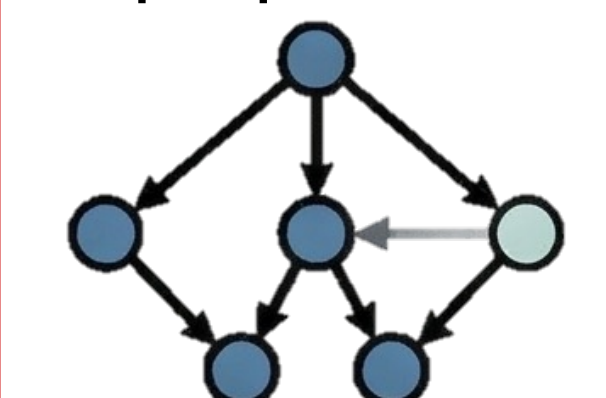
- **Shared Pipeline:** Data → Constraint Based Search → Score-based Optimization
- **Metrics:** BIC, Runtime, Prediction Accuracy

THE THREE HYBRID LEARNING ALGORITHMS

Phase1 :
Graph Skeleton Selection



Phase2:
Graph Optimization



MMHC(Max-Min Hill-Climbing)

MMPC (Constraint-based) + Hill-Climbing(score-based)

H2PC

HPC(Hybrid Parents & Children) + Hill-Climbing optimization

ARGES-like Pipeline

Adaptively Restricted GES using an HPC restriction graph

HEAD-TO-HEAD RESULTS

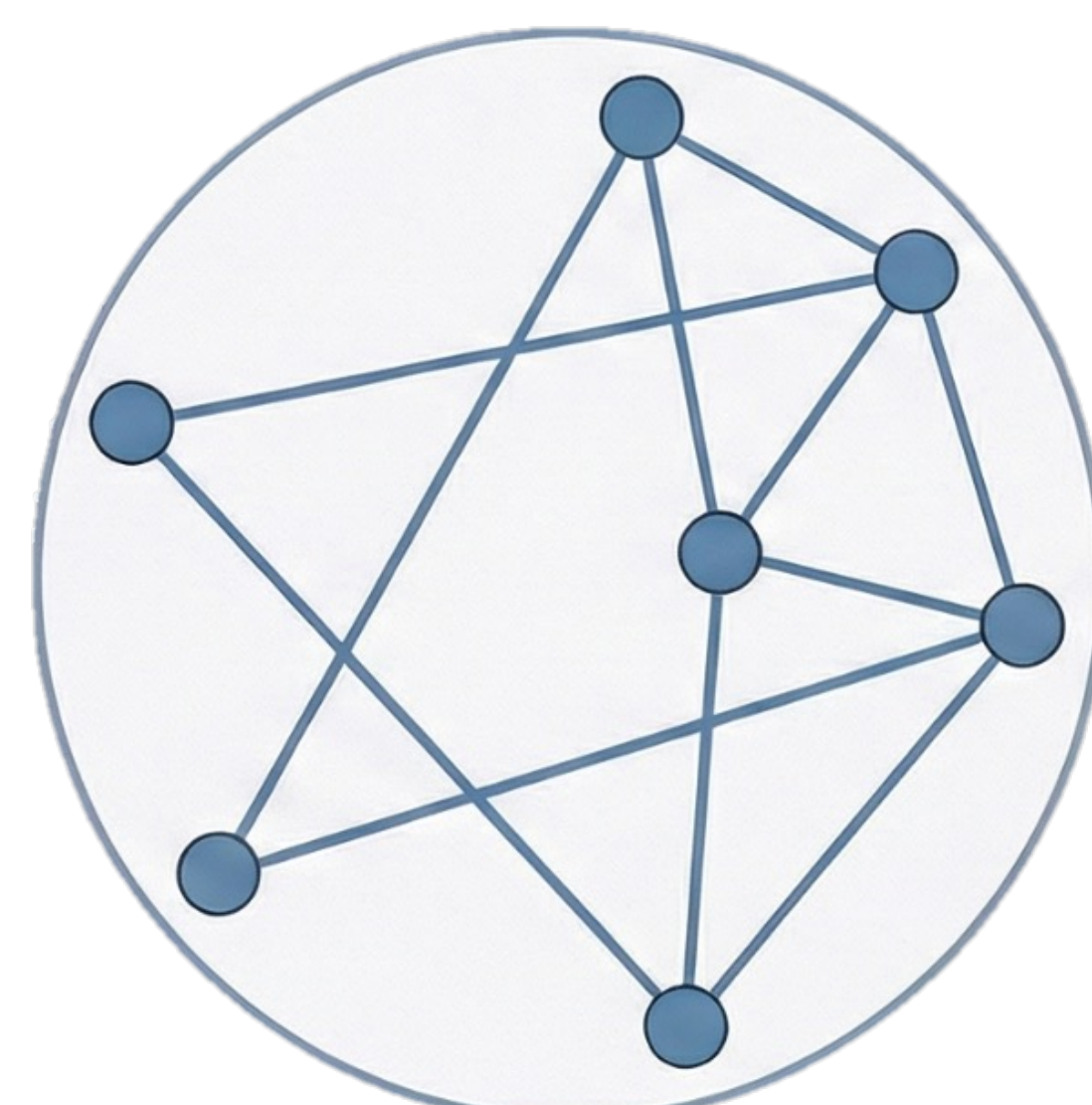
METHOD	BIC(Model Fit)	TOTAL RUNTIME	EDGE COUNT	PREDICTION ACCURACY	AUC
MMHC	-1,528,163.0048	16.9972 Maximum Speed	7	0.7044	0.7047
H2PC	-1,380,635.0229 BEST MODEL FIT	23.9098	25	0.7226 Highest Prediction	0.7922 Highest AUC
ARGES-like	-1,703,922.1521	21.4393	0 Failed to build	0.5595	0.5000

H2PC achieved the highest **predictive accuracy (0.7226)** while maintaining a rich and interpretable structure with **25 edges**. MMHC was **faster** but produced an **overly sparse** network, limiting clinical insight. **ARGES-like failed** to learn meaningful dependencies, suggesting it is not suitable for **categorical medical datasets**

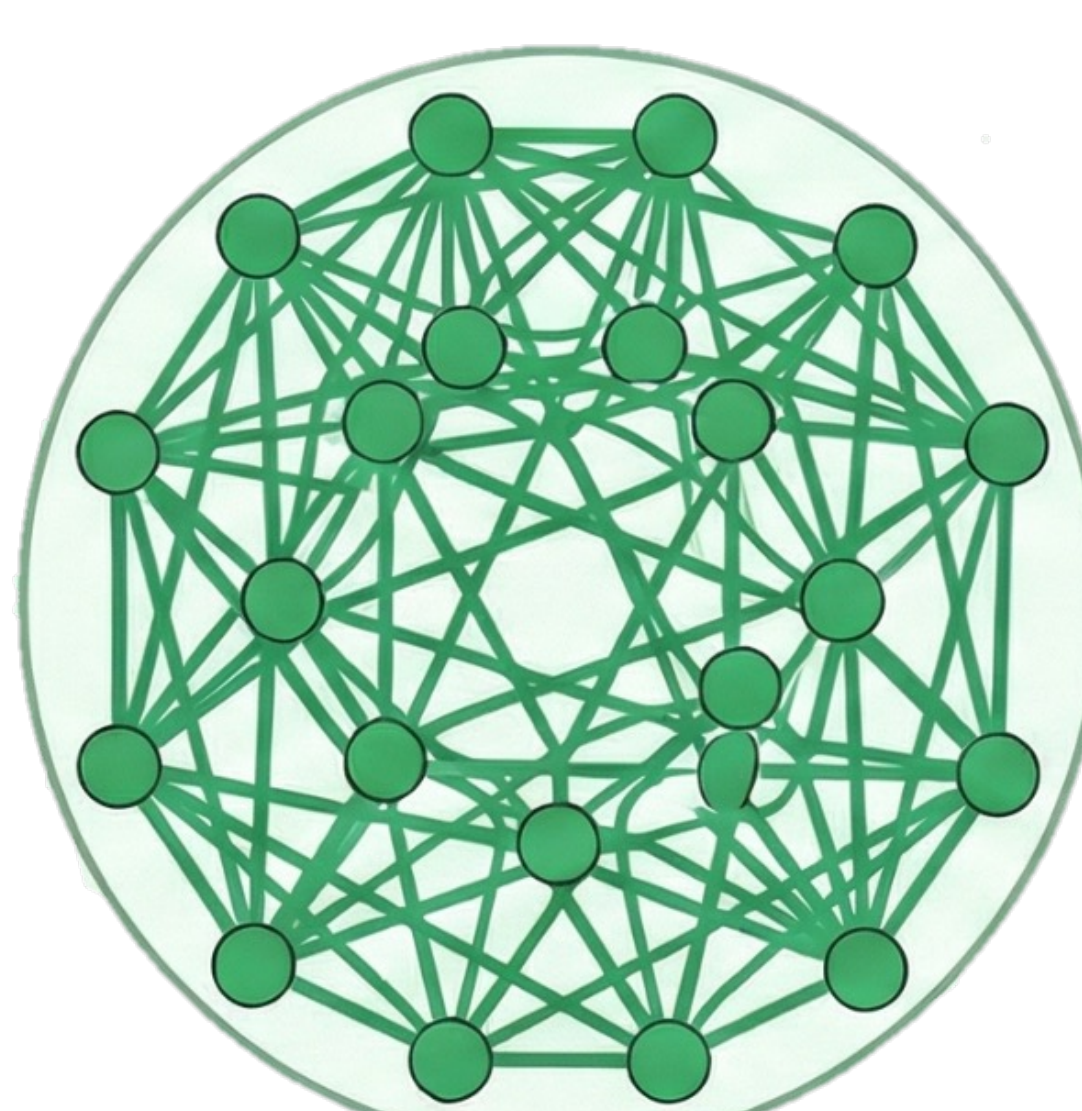
MMHC
Sparse – 7 edges

H2PC
Rich – 25 edges

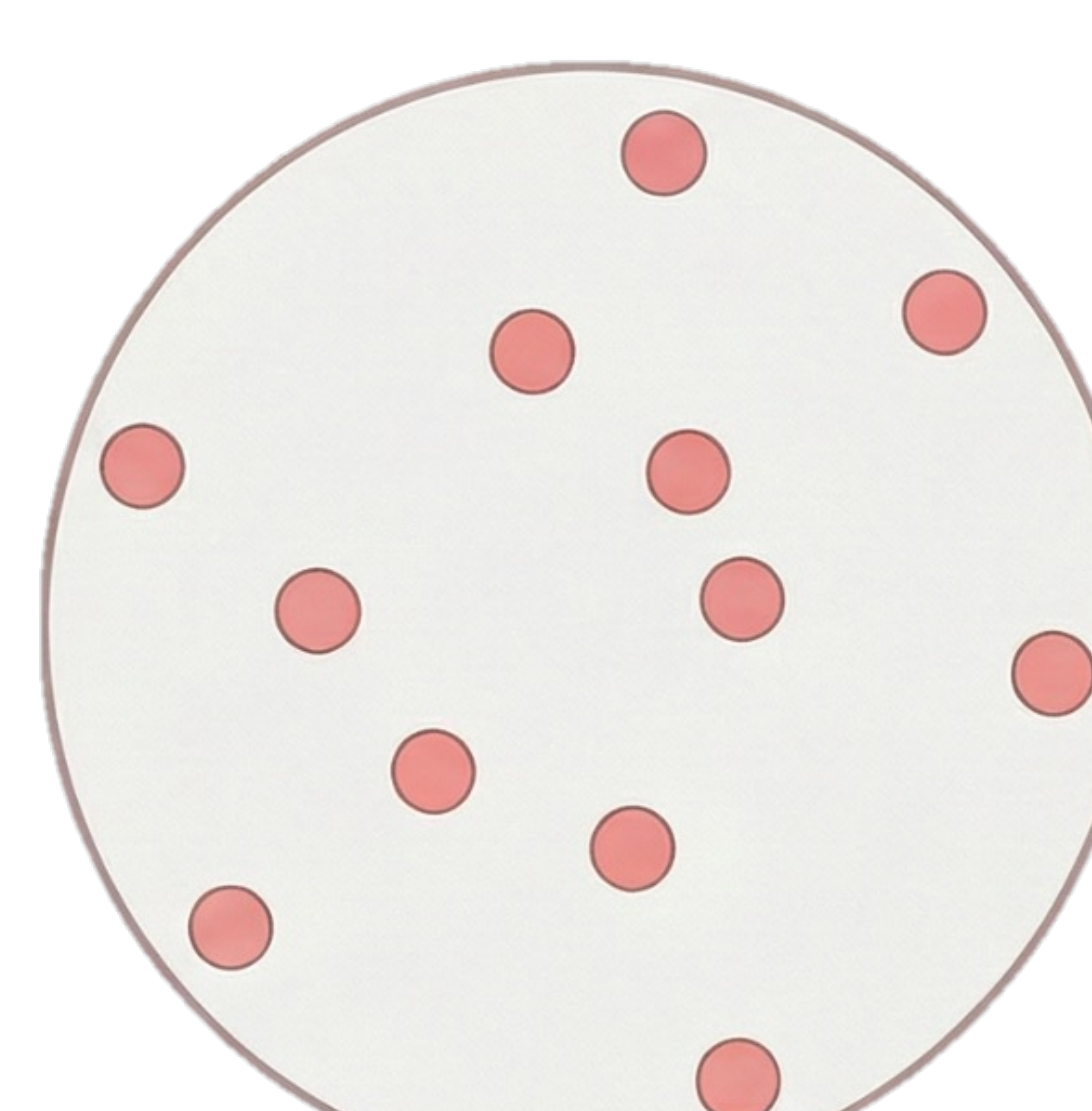
ARGES-like
No edge



Limited Number of dependencies identified

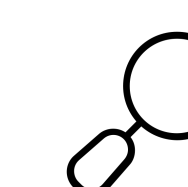


Identified many meaningful clinical edges



Failed to identify any meaningful edges

IMPACT



This work improves the reliability of Bayesian network models for clinical decision support in lung cancer.

By identifying structure-learning methods that produce both **accurate** and **interpretable models**, this research supports the development of **AI systems** that can assist clinicians in understanding **disease progression** and **treatment outcomes**. This is particularly valuable in **data-limited** or **underserved healthcare** settings where interpretability is essential for trust and adoption

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